ISyE 4600 - Fall 2025
Project Proposal

Team Member: Yichieh Chen

Project Title: Hybrid-ML Digital Twin for Real-Time Yield and Supply Chain Risk Prediction in Biopharmaceuticals

Abstract

This project proposes a Hybrid-ML Digital Twin framework designed to bridge the gap between real-time bioreactor telemetry and downstream supply chain logistics. By integrating mechanistic Monod kinetics with stochastic machine learning forecasting, the system enables proactive inventory optimization. The framework focuses on minimizing expected total costs across predicted yield distributions, effectively mitigating the 14-day-delay from biopharmaceutical manufacturing to the supply chain.

1 Problem Statement

Challenges in Bioprocess Digitalization
Biopharmaceutical manufacturing is navigating a "Data-Rich, Information-Poor" paradox. While bioreactors generate high-dimensional sensor data, the inherent biological stochasticity and complex metabolic interactions make accurate real-time yield prediction exceptionally difficult. Although computational methods have revolutionized protein structure prediction, AI-powered advancements in manufacturing processes and real-time quality control remain nascent.

The Mechanistic vs. Data-Driven Gap
Traditional mechanistic models, such as Monod kinetics, provide interpretability and adhere to fundamental mass balances; however, they often fail to account for the stochastic "noise" found in living systems. Conversely, pure Machine Learning (ML) models often perform poorly because they lack physical grounding, leading to "black-box" predictions that may violate thermodynamic constraints. Consequently, a *Hybrid-ML* approach is optimal for a "Quality by Design" (QbD) framework.

Mitigating Supply Chain Decoupling
As biologics are life-critical products, any latency in supply chain responsiveness is catastrophic. Currently, a systemic decoupling exists between manufacturing floor execution and inventory logistics. Because significant yield deficits are often identified only at the conclusion of a 14-day fermentation cycle, responses remain reactive. This project develops a *Digital Twin* that converts real-time telemetry into proactive supply chain intelligence.

2 Data Source

The project leverages a multi-fidelity data strategy to bridge the gap between static metabolic snapshots and dynamic process telemetry:

- **Historical Metabolic Dataset:** The Oyetunde et al. (2019) dataset provides 1,209 experimental observations. This includes metabolic features such as `atp_cost`, `nadh_nadph_cost`, molecular weights (`mw`), and genetic markers.
- **Real-Time Sensor Streams:** Publicly available Kaggle bioreactor telemetry data is utilized for live process simulation (pH, DO, Temp, Agitation).
  Dataset Source: `https://www.kaggle.com/datasets/cbeurrier/bioprocess-performance`

3 Methodology: The Hybrid-ML Framework

The Digital Twin is engineered as a three-layer hierarchical architecture:

3.1 Mechanistic ODE Layer

We define the biomass growth rate (μ) and substrate consumption (S) using the Monod relationship:

$$\frac{dX}{dt} = \mu(S)X - DX, \quad \mu(S) = \mu_{max}\frac{S}{K_s + S}$$

This layer enforces mass balance constraints, ensuring that predictions do not exceed theoretical stoichiometric maximums derived from `atp_cost`.

3.2 Soft-Sensor ML Layer

An XGBoost-based **Soft-Sensor** maps telemetry to latent Critical Quality Attributes (CQAs). The hybrid prediction is formulated as:

$$\text{Yield}_{pred} = Y_{mechanistic}(\text{atp}, \text{nadh}) + \mathcal{F}_{ML}(\text{pH}, \text{DO}, \text{Temp})$$

SHAP analysis will quantify the marginal contribution of each telemetry signal, enabling root-cause analysis when the Digital Twin detects a potential yield deficit.

3.3 Integration of Stochastic Forecasting and Supply Chain Optimization

The transition to the Supply Chain layer is achieved through Conditional Density Estimation. The model outputs a time-indexed probability density function (PDF) of the yield:

$$\hat{f}_t(y) = P(\text{Yield}_t = y \mid \text{Telemetry}_{1:t}, \text{Metabolic Features}) \tag{1}$$

The optimization framework utilizes this density to minimize the Total Expected Cost. The objective function integrates holding costs, backorder penalties, and proactive expedited sourcing costs (ΔP_t):

$$\min_{I_t, B_t, \Delta P_t} \sum_{t=1}^{T} \int_0^{\infty} \left[C_h \cdot I_t(y) + C_b \cdot B_t(y) + C_p \cdot \Delta P_t\right] \cdot \hat{f}_t(y)\, dy \tag{2}$$

Subject to stochastic flow constraints:

- **Inventory Balance:** $I_t(y) - B_t(y) = I_{t-1} + y + \Delta P_t - \text{Demand}_t$

4 Evaluation and Expected Impact

The project's performance will be validated through:

1. **Predictive Accuracy:** Target $R^2 > 0.85$ for the hybrid model vs. pure-ML baselines.
2. **Lead-Time Advantage:** Quantify the "Warning Window" between anomaly detection and batch completion.
3. **Economic Value Simulation:** A Monte Carlo analysis comparing the proposed system against a static inventory policy to measure reduction in total expected cost.